

# Session Objectives

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- 1) Understand how to evaluate memory loss and compare dementia to delirium.
- 2) Describe proper history taking in the setting of memory loss.
- 3) Generate a differential diagnosis for memory loss.
- 4) Consider reversible causes of memory loss.
- 5) Apply appropriate treatment options (if they exist).

# Take a good history

- Details:
- 1) Time course- slow or rapid (years or months?)
- 2) Trauma?
- 3) Fever or infection?
- 4) Toxic habits? HIV risk factors?
- 5) Vascular risk factors?
- 6) Other illnesses?

# Dementia

- Follows gradual course- months to years.
- With or without confusion, but disturbance in level of consciousness is generally not a feature.
- Dementia is an independent risk factor for delirium.

Diagnostic and Statistical Manual of Mental Disorders- 5th Edition  
Image: Official portrait

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# Irreversible Causes of Dementia

- Early or late onset Alzheimer's Disease
- Frontotemporal Dementia
- Chronic traumatic encephalopathy
- Lewy Body Dementias
- Vascular Cognitive Impairment
- Cerebral Amyloid Angiopathy
- Creutzfeld-Jakob, Prion disease

# Reversible Causes of Dementia

- Hypothyroidism
- Vitamin B1 or B12 Deficiency
- Depression
- Seizures
- Normal Pressure Hydrocephalus
- Subdural Hematoma
- Infection
- Brain Tumor
- Intracranial hypotension
- Medications, alcohol or drug dependency

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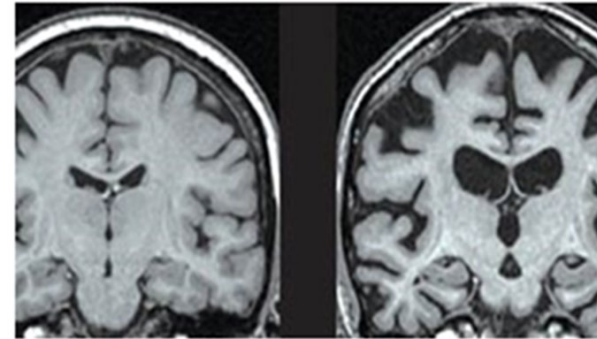
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## Clinical presentation (cognitive)

- Changes in verbal and nonverbal memory (asking the same questions and misplacing items)
- There may be difficulties with driving a car or managing complex paperwork.
- MoCA (Montreal Cognitive Assessment) is more sensitive and specific than MMSE in detecting the earliest stages of cognitive impairment.

# Imaging

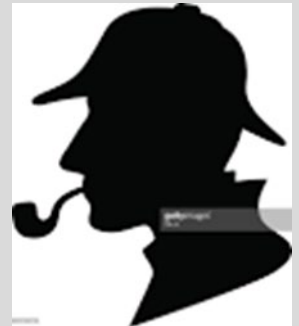
- Neuroimaging is used to exclude structural causes of dementia- including large strokes, brain tumors and infectious causes.
- MRI: progressive increased hippocampal volume loss over time.
- FDG-PET: glucose metabolism is measured with hypometabolism seen in the tempoparietal region in AD



# Mystery Case

- Over the following months Alfredo developed tremors, impotence and severe constipation.
- There were fevers of unclear cause.
- Blood work and imaging were all unrevealing.

The family that Couldn't Sleep DT MAX





## Case 2

- 70-year-old woman with hypertension referred for evaluation of memory loss.
- Cognitive changes for 3 years prior to presentation (unable to manage accounts at work and lost driving).
- Transient focal neurologic episodes (<30 minutes): motor, sensory, language or visual function- (AKA “amyloid spells”).
  - Paresthesias of the mouth, spreads to other body parts.
- MRI-numerous microhemorrhages

# Vascular Cognitive Impairment and Dementia

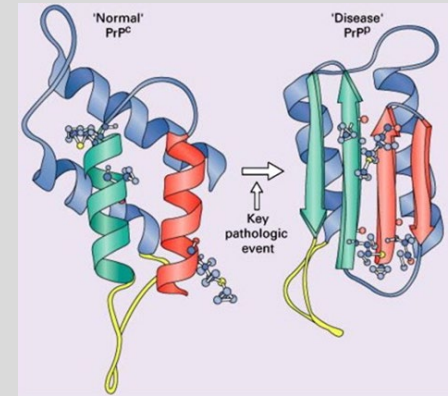
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- **Four Phenotypic Categories**
  - Subcortical ischemic vascular dementia- slowed processing speed.
  - Poststroke dementia- history of stroke with language or memory difficulties.
  - Multi-infarct dementia-Multiple large cortical infarcts with apraxia, aphasia, visual field cut or neglect
  - Mixed dementia- vascular cognitive impairment and AD: primarily amnestic syndrome.

# Prion Disease

- **Conversion** of the **normal prion protein (PrP<sup>c</sup>** prion related protein- c is the cellular form of the protein) It has a helical structure.
- An abnormal form of the protein- called the prion (PrP<sup>sc</sup> is the final product). Sc is for scrapie, the prion disease of sheep and goats).
- It has a B-pleated sheet structure.



# Revised Criteria for CJD

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## Core clinical features:

- Rapidly progressive cognitive impairment with 2 or more of the following:
  - Myoclonus
  - Visual or cerebellar disturbance
  - Pyramidal or extrapyramidal signs
  - Akinetic Mutism
  - PLUS imaging and biomarkers

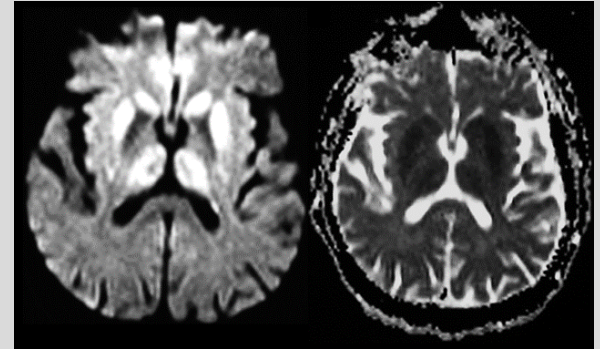


# Sporadic Creutzfeldt- Jakob

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- Spongiform Encephalopathy-fatal within one year.
- Sporadic form (85-95%)
- Age of onset: Seventh decade
- MRI- characteristic pattern
- EEG: generalized sharp or sharp-wave complexes
- “Positive” real-time quaking-induced conversion (RT-QuIC) assay for misfolded prion protein and elevated tau ( $>1150$ ) in the CSF.



## Case 4

- 74-year-old man is brought by his spouse
- “You are pretending to be your sister!”
- Significant cognitive “fluctuations”.
- Some days are lucid, other days not. On exam the patient appears parkinsonian
- He has visual hallucinations and is confused.

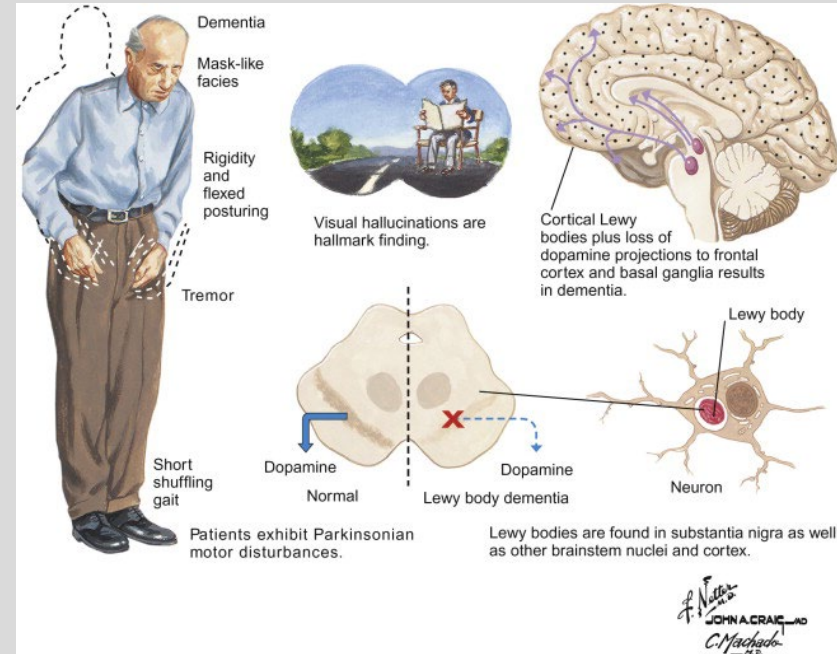


# Lewy Body Disease

- Fluctuating cognition or level of consciousness
- Visual hallucinations
- Parkinsonian motor signs
- REM sleep behavior disorder
- Syncope/postural instability
- Apathy, anxiety and depression

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# Lewy Body Disease

- Pathology: Intracytoplasmic neuronal inclusions.
- Cortically localized Lewy bodies are strongly related to cognitive impairment and dementia.





# Treatment for LBD

- Treatment is symptomatic only.
- Query family regarding safety concerns and driving.
- Sensitivity to meds is common.
- Identify medications that could contribute to clinical worsening- i.e. benzos and anti-cholinergics.
- Cholinesterase inhibitors treat cognitive symptoms and may help with behavioral symptoms.
- Typical antipsychotics are contraindicated.

# REM sleep behavior disorder

- Seen in LBD- may occur years before.
- In a healthy physiologic state- vivid mentation is accompanied by skeletal atonia.
- The loss of atonia during REM sleep can result in injurious dream enactment to the patient and/or bed partner.

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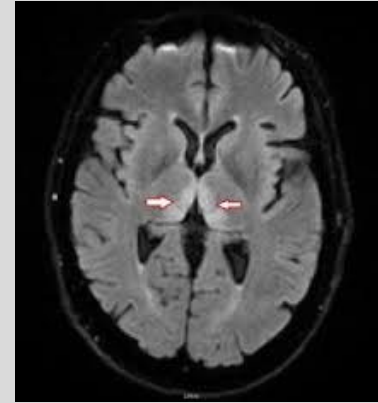


Image: Rembehaviorbiopsych.weebly.com. Lifelong Learning in Neurology. Continuum. Sleep Neurology. August 2020, Vol.26, No.4

# Wernicke's Encephalopathy

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- Severe thiamine deficiency (vitamin B1).
- Alcoholics or malnourished patients.
- Classic triad: global confusion, ophthalmoplegia (nystagmus) and ataxia.
- The classic triad not always present. Always have a high level of suspicion.



# Mystery Case

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- After several months Alfredo began to develop memory loss.
- He was also acting out activities during sleep in a mindless stupor.
- Hallucinations continued. He didn't know if he was alive or dead.

The Family that couldn't sleep- DT MAX



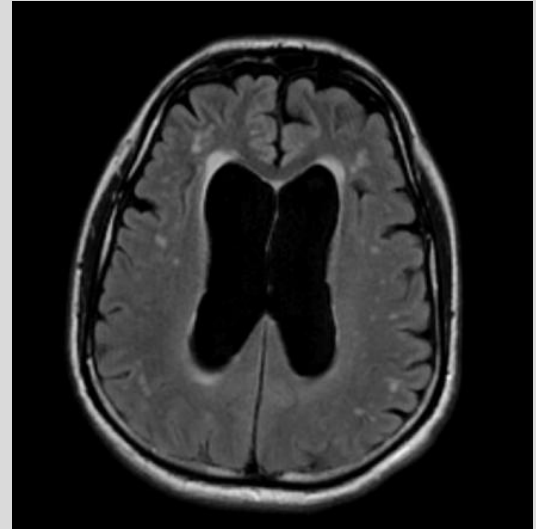
## Case 7

- 67-year-old woman with urinary incontinence, gait difficulty and memory loss (classic triad: wet, wobbly, wacky).
- Magnetic gait and appears parkinsonian.
- CT was performed and revealed large ventricles.

Case files of A. Leder, DO  
Image: Radiopaedia

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# Normal Pressure Hydrocephalus

- Potentially reversible form of memory loss and parkinsonism.
- Gait disturbance and incontinence due to pressure on and distortion of descending white matter pathways.
- Cognitive symptoms due to pressure on anterior cortical structures.

# NPH- Treatment

- Videotape gait before the LP
- 30 ml of CSF is drained.
- 30 minutes after LP- observe patient walking again.
- Measure gait, cognition, balance and continence based on specialized scales.
- Determine if VP shunt is the appropriate treatment.

## Case 8

- 51-year-old woman –evaluation for 4 years of behavioral symptoms.
- Was a business executive. Interjected political views, belittled clothing or body habitus of staff members.
- Preoccupied with sugary drinks.
- Starting collecting antique statuettes (\$15,000).
- Refused to drive kids to their games “teams stink anyway”.
- Gifted partly used cigar (found on street) for Father’s Day.

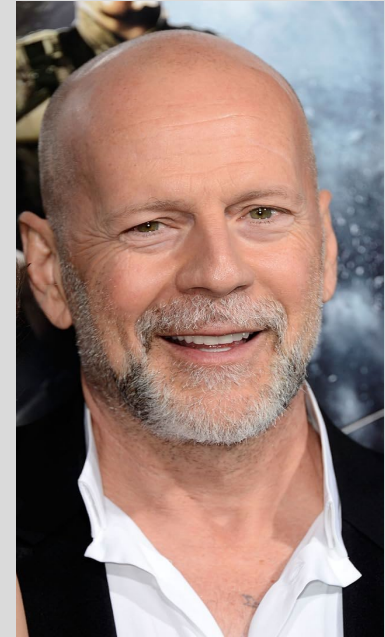


# Frontotemporal Dementia

- Specific syndromes under the FTD umbrella include behavior behavioral variant (bvFTD) and primary progressive aphasia (PPA).
- Most cases are sporadic. Familial phenotypes are associated with the *MAPT* gene, *GRN* gene and the *C9orf72* gene.

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# Frontotemporal Dementia

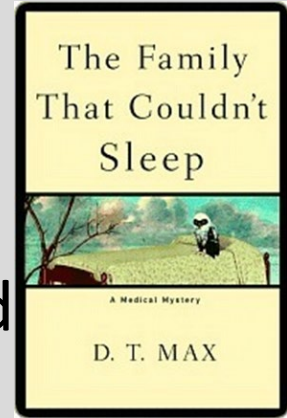
- Diagnosis: Functional neuroimaging (Fludeoxyglucose PET).
- No FDA approved medication at this time.
- Symptoms treated with SSRIs, atypical antipsychotics and trazodone for behavioral dyscontrol and agitation.
- Cholinesterase inhibitors do not improve cognition.

# Mystery Case- Revealed!

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- Fatal familial insomnia
- Rare genetic prion disease
- Onset in late 40s
- Severe progressive insomnia over months
- Dysautonomia (tachycardia, hyperhidrosis, and hyperpyrexia).
- Motor and cognitive symptoms occur late.
- 18 months survival



The family that couldn't sleep- DT MAX